
Problem Set 5 - Solutions

Problem 3.44

Calculate Z and V for ethylene at 25°C and 12 bar by the following equations:

(a) The truncated virial equation (Eq. 3.38) with the following experimental values of virial coefficients:

$$B = -140. \frac{\text{cm}^3}{\text{mol}} \text{ and } C = 7,200 \frac{\text{cm}^6}{\text{mol}^2}$$

(b) The truncated virial equation (Eq. 3.36), with a value of B from the generalized Pitzer correlation (Eqs. 3.58-3.62)

(c) The Redlich/Kwong equation.

(d) The Soave/Redlich/Kwong equation.

(e) The Peng/Robinson equation.

Solution to Part (a) - Truncated virial equation (Eq. 3.38)

```
In[1]:= R = 83.14 (* cm^3*bar / (mol*K) *);  
P = 12. (* bar *);  
T = 298.15 (* K *);  
B = -140. (* cm^3 / mol *);  
C = 7200. (* cm^6 / mol^2 *);  
  
In[6]:= (P * V) / (R * T) == 1 + (B / V) + (C / V^2) // Solve // Quiet (*Eq 3.38*)  
Out[6]= {{V -> 73.33041 - 48.71252 I}, {V -> 73.33041 + 48.71252 I}, {V -> 1919.022}}
```

$$Z = \frac{P \cdot V}{R \cdot T}$$

```
In[7]:= Z  
Out[7]= 0.9290014  
  
(*V=1919.022 cm^3/mol and Z=0.9290 //ANS Part (a)*)
```

Solution to Part (b) - Truncated virial equation (Eq. 3.36) with B from Eqns. 3.58-3.62.

```
In[8]:= (*Table B.1*)
Tc = 282.3; (*K*)
Pc = 50.4; (*bar*)
ω = 0.087;

Tr = T / Tc;

B0 = 0.083 -  $\frac{.422}{Tr^{1.6}}$ ; (*Eq. 3.61*)
B1 = 0.139 -  $\frac{0.172}{Tr^{4.2}}$ ; (*Eq. 3.62*)

B̂ = B0 + ω * B1; (*Eq. 3.59*)
B =  $\frac{\hat{B} * R * Tc}{Pc}$  (*Eq. 3.58*);

In[16]:=  $\frac{P * V}{R * T} == 1 + \frac{B * P}{R * T}$  // Solve // Quiet (*Eq. 3.36*)
Out[16]:= {{V → 1924.355}}

In[17]:=  $\frac{P * 1924.355}{R * T}$ 
Out[17]:= 0.9315831

(*V=1924.355  $\frac{cm^3}{mol}$  and Z=0.9316 //ANS Part (b)*)
```

Solution to Part (c) - Redlich/Kwong equation

```
In[18]:= (*Table 3.1 for RK equation*)
Ω = 0.08664;
Ψ = 0.42748;
α = (Tr)-1/2;
σ = 1;
ε = 0;

a = Ψ  $\frac{\alpha * R^2 * Tc^2}{Pc}$ ; (*Eq 3.45*)
b = Ω *  $\frac{R * Tc}{Pc}$ ; (*Eq 3.44*)
```

```

In[25]:= P ==  $\frac{R * T}{V - b} - \frac{a}{(V + \epsilon * b) * (V + \sigma * b)}$  // Solve // Quiet(*Eq 3.41*)
Out[25]=
{ {V → 74.59371 - 49.10982 i}, {V → 74.59371 + 49.10982 i}, {V → 1916.495} }

In[26]:=  $\frac{P * 1916.495}{R * T}$  (*=Z*)
Out[26]=
0.9277781

(*V=1916.495  $\frac{\text{cm}^3}{\text{mol}}$  and Z=0.9278 //ANS Part (c)*)

```

Solution to Part (d) - Soave/Redlich/Kwong equation

```

(*Table 3.1, SRK - NEW stuff only*)
 $\alpha = (1 + (0.480 + 1.574 * \omega - 0.176 * \omega^2) * (1 - \sqrt{T_r}))^2;$ 

 $a = \Psi \frac{\alpha * R^2 * T_c^2}{P_c}$  (*Eq 3.45, do not need to recalculate b*);

In[29]:= P ==  $\frac{R * T}{V - b} - \frac{a}{(V + \epsilon * b) * (V + \sigma * b)}$  // Solve // Quiet(*Eq 3.41*)
Out[29]=
{ {V → 73.85654 - 49.58738 i}, {V → 73.85654 + 49.58738 i}, {V → 1917.969} }

In[30]:=  $\frac{P * 1917.969}{R * T}$  (*=Z*)
Out[30]=
0.9284916

(*V=1917.969  $\frac{\text{cm}^3}{\text{mol}}$  and Z=0.9285 //ANS Part (d)*)

```

Solution to Part (e) - Peng/Robinson

```

In[38]:= (*Table 3.1, PR*)
 $\Omega = 0.0778;$ 
 $\Psi = 0.45724;$ 
 $\alpha = (1 + (0.37464 + 1.54226 * \omega - 0.26992 * \omega^2) * (1 - T_r^{1/2}))^2;$ 
 $\epsilon = 1 - \sqrt{2};$ 
 $\sigma = 1 + \sqrt{2};$ 

 $a = \Psi \frac{\alpha * R^2 * T_c^2}{P_c}$  (*Eq 3.45*);
 $b = \Omega * \frac{R * T_c}{P_c}$  (*Eq 3.44*);

```

```

In[45]:= P == 
$$\frac{R * T}{V - b} - \frac{a}{(V + \epsilon * b) * (V + \sigma * b)}$$
 // Solve // Quiet (*Eq 3.41*)
Out[45]=
{{V -> 64.44718 - 45.96281 i}, {V -> 64.44718 + 45.96281 i}, {V -> 1900.558}}

In[46]:= 
$$\frac{P * 1900.558}{R * T} (*=Z*)$$

Out[46]=
0.9200629

(*V=1900.558  $\frac{\text{cm}^3}{\text{mol}}$  and Z=0.9201 //ANS Part (e)*)

```

Problem 3.58

To a good approximation, what is the molar volume of ethanol vapor at 480°C and 6,000 kPa? How does this result compare with the ideal gas value?

Compare the results from (a) the Lee-Kesler method, (b) SRK equation of state, and (c) ideal gas equation of state.

Solution to Part (a) - Lee-Kesler

```

In[47]:= R = 83.14 ; (*  $\frac{\text{cm}^3 * \text{bar}}{\text{mol} * \text{K}}$  *)

(*Table B.1 lookups, page 664, for ethanol*)
omega = 0.645; (*dimensionless*)
Tc = 513.9; (*K*)
Pc = 61.48 ; (*bar*)

P = 60.; (*bar given 6000 kPa with 1 bar = 100 kPa*)
T = 273.15 + 480 ; (*K*)

Tr = T / Tc
Pr = P / Pc

Out[53]=
1.465558

Out[54]=
0.9759271

(*Use these values with Tables D.1 and D.2 and online interpolator to get Z0 and Z1.*)

In[55]:= Z0 = .903149;
Z1 = .083871;
Z = Z0 + omega * Z1

Out[57]=
0.9572458

```

```
In[58]:= Va =  $\frac{Z * R * T}{P}$ 
```

```
Out[58]= 998.9959
```

```
(*V=999.00  $\frac{\text{cm}^3}{\text{mol}}$  //ANS Part (a) *)
```

Solution to Part (b) - Soave/Redlich/Kwong

```
(*Table 3.1 lookups, page 100, for SRK*)
```

```
 $\Omega = 0.08664;$ 
```

```
 $\Psi = 0.42748;$ 
```

```
 $\sigma = 1;$ 
```

```
 $\epsilon = 0;$ 
```

```
 $\alpha = (1 + (0.480 + 1.574 * \omega - 0.176 * \omega^2) * (1 - \sqrt{T_r}))^2;$ 
```

```
 $b = \Omega * \frac{R * T_c}{P_c} (*Eq 3.44*);$ 
```

```
 $a = \Psi * \frac{\alpha * R^2 * T_c^2}{P_c} (*Eq 3.45*);$ 
```

```
In[66]:= P ==  $\frac{R * T}{V - b} - \frac{a}{(V + \epsilon * b) * (V + \sigma * b)}$  // Solve // Quiet(*Eq 3.41*)
```

```
Out[66]= {{V -> 15.3922 - 77.03595 I}, {V -> 15.3922 + 77.03595 I}, {V -> 1012.83}}
```

```
(*V=1012.83  $\frac{\text{cm}^3}{\text{mol}}$  //ANS Part (b) *)
```

Solution to Part (c) - Ideal Gas

```
In[67]:= eqb =  $\frac{P * V}{R * T} == 1;$ 
```

```
Quiet[solb = Solve[eqb, V]]
```

```
Out[68]= {{V -> 1043.615}}
```

```
(*V=1043.61  $\frac{\text{cm}^3}{\text{mol}}$  //ANS Part (b) *)
```

Comparison

In an ideal gas, according to the kinetic molecular theory, there are assumed to be no intermolecular forces of attraction (IMFs) and the molecules themselves occupy zero volume. That is, in the ideal gas equation of state, as $T \rightarrow 0$, $V \rightarrow 0$. In a real gas, on the other hand, the IMFs collectively “pull” on the molecules and the models are corrected for nonzero molecular volume. As a molecule approaches the wall of the container, there is a collective pull from the IMFs of the other molecules in the bulk fluid. This average inward pull results in a lower molar volume (at the same pressure). //ANS

INSTRUCTOR'S NOTE (not needed for cadet submissions)

Consider that as T is dropped to the point where condensation is observed, the liquid volume is on the order of 30-50 cm³/mol (see for example 3.14 in lesson 14 slides 5-7 for a calculation).

Problem 3.78

The *Boyle temperature* is the temperature for which:

$$\lim_{P \rightarrow 0} \left(\frac{\partial Z}{\partial P} \right)_T = 0$$

(a) Show that the second virial coefficient B is zero at the Boyle temperature.

(b) Use the generalized correlation for B, Eqs. 3.58 to 3.62, to estimate the reduced Boyle temperature for simple fluids.

Solution to Part (a)

```
In[71]:= ClearAll["Global`*"]
```

Start by taking the limit of $(\partial Z / \partial P)_T$ as P approaches 0:

```
In[72]:= Z = 1 + B' * P + C' * P^2 + D' * P^3 (*Lesson 13, Eq. 3.33, truncated to four terms*);
```

```
TraditionalForm[HoldForm[Limit[∂PZ, P → 0]] == Limit[∂PZ, P → 0]]
```

```
Out[73]//TraditionalForm=
```

$$\lim_{P \rightarrow 0} \frac{\partial Z}{\partial P} = B'$$

Then apply the definition of the Boyle Temperature:

```
In[74]:= TraditionalForm[HoldForm[Limit[∂PZ, P → 0]] == 0]
```

```
Out[74]//TraditionalForm=
```

$$\lim_{P \rightarrow 0} \frac{\partial Z}{\partial P} = 0$$

These calculations show that the limit of $(\partial Z / \partial P)_T$ as P approaches 0 is B' , and that this limit is also equal to zero. Therefore, $B' = 0$ at $T = T_{\text{Boyle}}$.

However, equation 3.35a states that $B' = \frac{B}{R \cdot T}$, where R is a positive constant. Therefore, for any nonzero value of T, since $B' = 0$, then B must also be zero. //ANS

Solution to Part (b)

The acentric factor ω is zero for simple fluids by definition. (Simple fluids include Argon, Krypton, and Xenon.) Equation 3.59, the Pitzer equation for the reduced second virial coefficient $\hat{B}$, and states that $\hat{B} = B^0 + \omega B^1$. If $\omega=0$, then $\hat{B} = B^0$.

Equation 3.58 defines the reduced second virial coefficient $\hat{B}$ in terms of B as $\hat{B} = \frac{BP_c}{RT_c}$. If $B=0$ (from part (a)), then $\hat{B}=0$ and $B^0=0$.

Equation 3.61 states that $B^0 = 0.083 - \frac{0.422}{T_r^{1.6}} = 0$, which can be solved for T_r :

```
In[75]:= 0 == 0.083 -  $\frac{0.422}{T_r^{1.6}}$  // Solve // Quiet
```

```
Out[75]= {{Tr → 2.763 099}}
```

Therefore the reduced temperature is 2.763 for the simple fluids at the Boyle temperature. //ANS

INSTRUCTOR'S NOTE (not needed for cadet submissions)

The actual Boyle temperatures in degrees K can be calculated using the critical temperatures from Appendix B as follows:

```
(*Simple fluids are Ar, Kr, Xe*)
(*Lookup Tc values in Table B.1*)
(*Temperatures are in degrees K*)
Tc = {150.9, 209.4, 289.7};
2.763 099 * Tc
```

```
Out[78]= {416.9516, 578.5929, 800.4698}
```